Lab Guide: Create and Configure VPC Networks

About this lab

Every Compute Engine VM, Kubernetes Engine container and App Engine flexible application needs a Virtual Private Cloud (VPC) network to run in. A VPC network works much like a physical network, but it lives entirely inside Google Cloud. It is a global resource made up of regional subnets, all linked by Google's global wide area network, and each VPC network is isolated from every other one.
In this lab you start by removing the default network to see what happens without one. You then build three networks of your own, add firewall rules and VMs, and test which VMs can reach each other.

What you will build

Networks and subnets:

Network	**Mode**	**Subnet**	**Region**	**IP range**
mynetwork	Auto, then converted to custom	One per region (created automatically)	All regions	Assigned automatically
managementnet	Custom	managementsubnet-us	Region 1	10.240.0.0/20
privatenet	Custom	privatesubnet-us	Region 1	172.16.0.0/24
privatenet	Custom	privatesubnet-notus	Region 2	172.20.0.0/20

Virtual machines:

VM	**Network**	**Location**	**Machine type**	**Created with**
mynet-us-vm	mynetwork	Region 1 / Zone 1	e2-medium	Cloud console
mynet-notus-vm	mynetwork	Region 2 / Zone 2	e2-medium	Cloud console
managementnet-us-vm	managementnet	Region 1 / Zone 1	e2-micro	Cloud console
privatenet-us-vm	privatenet	Region 1 / Zone 1	e2-micro	gcloud

About region and zone names: this guide uses Region 1, Zone 1, Region 2 and Zone 2. Your Lab setup and access panel gives the actual values to use. In commands, replace [REGION_1], [ZONE_1] and [REGION_2] with those values.

What you will learn

- What the default VPC network contains and what happens without any network
- How to create an auto mode network with firewall rules, and convert it to custom mode
- How to create custom mode networks and firewall rules with both the console and gcloud
- How network membership, not location, decides whether VMs can talk over internal IP addresses

Before you begin: open the lab environment

1. Select **Start Lab**. If the lab requires payment, choose a payment method in the pop-up.
2. Check the **Lab setup and access** panel for your temporary username and password, the **Open Google Cloud console** button, and your regions and zones. The timer at the top shows how long you have left.
3. Select **Open Google Cloud console**. In Chrome, you can right-click and open it in an Incognito window instead.
4. If asked to pick an account, choose **Use Another Account**.
5. Enter the lab **Username** and select **Next**, then enter the lab **Password** and select **Next**.
6. Accept the terms. Skip recovery options, two-factor authentication and free-trial offers, since this account exists only for the lab.

Important: Use only the lab credentials. Signing in with your own Google Cloud account could lead to charges.

Task 1. Investigate the default network

A new project comes with a network called default, which already has subnets, routes and firewall rules. You'll look at each part, then delete the network to see what depends on it.

Subnets

1. Go to **Navigation menu** > **VPC network** > **VPC networks**.
2. Look at the default network. It has one subnet in every Google Cloud region. Each subnet has its own private RFC 1918 CIDR range for internal IP addresses, plus a gateway.

Routes

Routes decide where traffic leaving a VM goes, whether to another subnet or out to the internet.

1. In the left pane, select **Routes**.
2. Set **Network** to **default** and **Region** to **Region 1**, then select **View**.
3. You'll see one route per subnet, plus a route to the default internet gateway (0.0.0.0/0). Google manages these for you. You can also add custom static routes, for example to send outbound traffic through a VM acting as a NAT gateway.

Firewall rules

Each VPC network has a distributed virtual firewall. Two implied rules are always present: one blocks all incoming connections and one allows all outgoing connections. Other rules are layered on top.

1. In the left pane, select **Firewall**.
2. Find the four ingress rules on the default network, summarized below. The Targets, Filters, Protocols/ports and Action columns show how each rule works.

Rule	Allows	From
default-allow-icmp	ICMP (ping)	Anywhere (0.0.0.0/0)
default-allow-rdp	RDP	Anywhere (0.0.0.0/0)
default-allow-ssh	SSH	Anywhere (0.0.0.0/0)
default-allow-internal	All TCP, UDP and ICMP	Inside the network (10.128.0.0/9)

Remove the firewall rules and the network

1. In the left pane, select **Firewall policies**.
2. Under **VPC firewall rules**, select every rule that belongs to the default network, select **Delete**, and confirm.
3. Go to **VPC network** > **VPC networks** and open the **default** network.
4. Select **Delete VPC network**, type **default**, and select **Delete**. Wait until the deletion finishes.
5. Open **Routes** and then **Firewall** in the left pane. Both lists are now empty: with no network, there are no routes or rules.

Try to create a VM without a network

1. Go to **Compute Engine** > **VM instances** and select **Create instance**.
2. Keep the defaults and select **Create**. An error appears on the **Networking** tab.
3. Select **Go to Issues**. Under **Network interfaces**, the message **No more networks available** confirms the VM can't be created.
4. Select **Cancel**.

Task 2. Build an auto mode network

An auto mode network creates a subnet in every region for you. That makes it quick for prototyping, but you give up control over which regions and IP ranges are used. You'll use one here, then convert it later.

Enable the APIs you need

1. Select **Activate Cloud Shell** in the console toolbar, and select **Continue** if prompted.

2. Enable the Identity-Aware Proxy and Network Management APIs:

```
gcloud services enable \
iap.googleapis.com \
networkmanagement.googleapis.com
```

Create mynetwork

1. Go to **VPC network** > **VPC networks** and select **Create VPC network**.
2. Name it **mynetwork** and set **Subnet creation mode** to **Automatic**.
3. Under **Firewall rules**, select every rule offered. These match the rules the default network had. The implied deny-all-ingress and allow-all-egress rules are shown too, but can't be changed. They have the lowest priority (a higher number means lower priority), so the allow rules are checked first.
4. Select **Create**.
5. When it's ready, open **mynetwork** > **Subnets** and note the IP ranges for Region 1 and Region 2.

Tip: If you ever delete a project's default network by mistake, creating an auto mode network like this one gives you an equivalent replacement.

Allow SSH through Identity-Aware Proxy

This rule lets you SSH into VMs through IAP, which connects from Google's 35.235.240.0/20 range, without exposing SSH more widely.

1. Go to **VPC network** > **Firewall** and select **Create firewall rule**.
2. Enter the settings below, then select **Create**.

Setting	**Value**
Name	allow-iap-ssh
Network	mynetwork
Priority	1000
Direction of traffic	Ingress
Action on match	Allow
Targets	Specified target tags
Target tags	iap-gce
Source filter	IPv4 ranges
Source IP ranges	35.235.240.0/20
Protocols and ports	Specified protocols and ports: tcp:22

Add two VMs to mynetwork

The region and zone you pick decide which subnet a VM joins and where its internal IP address

comes from. Create both VMs from **Compute Engine** > **VM instances** > **Create instance**, using these settings:

Setting	First VM	Second VM
Name	mynet-us-vm	mynet-notus-vm
Region / Zone	Region 1 / Zone 1	Region 2 / Zone 2
Series	E2	E2
Machine type	e2-medium (2 vCPU, 4 GB memory)	e2-medium (2 vCPU, 4 GB memory)
OS and storage	The Debian image your lab specifies	The Debian image your lab specifies
Networking > Network tags	iap-gce	iap-gce

For each VM: fill in the name, location and machine type; open **OS and storage** and select **Change** if the image isn't the Debian version your lab specifies (debian_image_name in the lab); add the network tag under **Networking**; then select **Create**.
About external IPs: Both VMs get ephemeral external IP addresses. If a VM is stopped, its ephemeral address goes back to the shared pool and it gets a new one when it starts again. To keep an address permanently, reserve a static external IP.

Test connectivity inside mynetwork

1. On the **VM instances** page, note the internal and external IP addresses of **mynet-notus-vm**.
2. In Cloud Shell, connect to mynet-us-vm through IAP, replacing [ZONE_1] with your Zone 1 value:

```
gcloud compute ssh mynet-us-vm \
--zone=[ZONE_1] \
--tunnel-through-iap
```

3. Enter **Y** to continue. When asked for a passphrase, press **ENTER** twice to leave it empty.
4. Ping mynet-notus-vm's internal IP address:

```
ping -c 3 [MYNET_NOTUS_INTERNAL_IP]
```

5. Ping mynet-notus-vm's external IP address:

```
ping -c 3 [MYNET_NOTUS_EXTERNAL_IP]
```

6. Type **exit** to close the SSH session.

Why this works: The SSH connection succeeds because of the allow-iap-ssh rule and the iap-gce tag on the VM. Compute Engine creates an SSH key for you and stores it in project or instance metadata, or with your user account if OS Login is enabled. If metadata updates are blocked, SSH can fail even when port 22 is open. The internal ping succeeds because of mynetwork-allow-custom, and the external ping because of mynetwork-allow-icmp. The same tests also work in the other direction, from mynet-notus-vm to mynet-us-vm.

Switch mynetwork to custom mode

Auto mode keeps adding subnets as Google opens new regions. Those new ranges could overlap with subnets or static routes you add yourself, so production networks should use custom mode.

1. Go to **VPC network** > **VPC networks** and open **mynetwork**.
2. Select **Edit**, set **Subnet creation mode** to **Custom**, and select **Save**.
3. Go back to the VPC networks list and wait for the **Mode** of mynetwork to show **Custom**. Select **Refresh** if needed.

Task 3. Build two custom mode networks

Now you add managementnet and privatenet, with firewall rules for SSH, ICMP and RDP, and one VM in each. Their IP ranges don't overlap with each other or with mynetwork.
Non-overlapping ranges make it possible to connect networks later with VPC peering, or to an on-premises network with VPN or Cloud Interconnect.

Create managementnet in the console

1. Go to **VPC network** > **VPC networks** and select **Create VPC Network**.
2. Name it **managementnet** and set **Subnet creation mode** to **Custom**.
3. Add a subnet named **managementsubnet-us** in **Region 1** with IPv4 range **10.240.0.0/20**, leave other settings at their defaults, and select **Done**.
4. Select **Equivalent Command line** to see the gcloud commands the console would run. You'll use the same pattern for privatenet. Select **Close**.
5. Select **Create**.

Create privatenet with gcloud

1. In Cloud Shell, create the network. Select **Authorize** if prompted.

```
gcloud compute networks create privatenet --subnet-mode=custom
```

2. Add the Region 1 subnet:

```
gcloud compute networks subnets create privatesubnet-us
--network=privatenet --region=[REGION_1] --range=172.16.0.0/24
```

3. Add the Region 2 subnet:

```
gcloud compute networks subnets create privatesubnet-notus
--network=privatenet --region=[REGION_2] --range=172.20.0.0/20
```

4. List your networks. You should see managementnet, mynetwork and privatenet, all with SUBNET_MODE: CUSTOM.

```
gcloud compute networks list
```

5. List the subnets, grouped by network:

```
gcloud compute networks subnets list --sort-by=NETWORK
```

6. In the console, open **VPC network** > **VPC networks** and confirm the same networks and subnets appear.

Note: managementnet and privatenet contain only the subnets you created. mynetwork has a subnet in every region because it started in auto mode, even though it's custom now.

Add a firewall rule to managementnet in the console

1. Go to **VPC network** > **Firewall** and select **Create firewall rule**.
2. Enter the settings below. Be sure to include /0 in the source range so the rule covers all addresses.

Setting	**Value**
Name	managementnet-allow-icmp-ssh-rdp
Network	managementnet
Targets	All instances in the network
Source filter	IPv4 Ranges
Source IPv4 ranges	0.0.0.0/0
Protocols and ports	Specified protocols and ports: tcp ports 22 and 3389, and Other protocols: icmp

3. Select **Equivalent Command line** to see the gcloud version, then **Close**.
4. Select **Create**.

Add a firewall rule to privatenet with gcloud

1. In Cloud Shell, create the rule:

```
gcloud compute firewall-rules create privatenet-allow-icmp-ssh-rdp
--direction=INGRESS --priority=1000 --network=privatenet
--action=ALLOW --rules=icmp,tcp:22,tcp:3389 --source-ranges=0.0.0.0/0
```

2. List all firewall rules, grouped by network:

```
gcloud compute firewall-rules list --sort-by=NETWORK
```

3. Check the output. managementnet and privatenet each have one rule covering several protocols at priority 1000. mynetwork has four separate rules (allow-custom, allow-icmp, allow-rdp, allow-ssh) at priority 65534. Both styles are valid.
4. In the console, open **VPC network** > **Firewall** and confirm the same rules appear.

Create managementnet-us-vm in the console

1. Go to **Compute Engine** > **VM instances** and select **Create instance**.
2. Enter the settings below.

Setting	**Value**
Name	managementnet-us-vm
Region / Zone	Region 1 / Zone 1
Series	E2
Machine type	e2-micro (2 vCPU, 1 core, 1 GB memory)
Boot disk	Debian image

Networking > Network	managementnet
Networking > Subnetwork	managementsubnet-us

3. To set the network, open **Networking**, expand the network interface, choose the network and subnetwork, and select **Done**. Only subnets in the selected region are offered.
4. Select **Equivalent Code** to see the matching gcloud command, then select **Create**.

Create privatenet-us-vm with gcloud

1. In Cloud Shell, create the VM. Replace [ZONE_1] with your Zone 1 value, and the image family and project placeholders with the values your lab gives (debian_image_id_short and debian_image_project in the lab):

```
gcloud compute instances create privatenet-us-vm --zone=[ZONE_1]
--machine-type=e2-micro --subnet=privatesubnet-us
--image-family=[DEBIAN_IMAGE_FAMILY]
--image-project=[DEBIAN_IMAGE_PROJECT] --boot-disk-size=10GB
--boot-disk-type=pd-standard --boot-disk-device-name=privatenet-us-vm
```

2. List all VMs by zone:

```
gcloud compute instances list --sort-by=ZONE
```

3. In the console, open **Compute Engine** > **VM instances**, confirm the VMs are listed, and add the **Zone** column.

Three VMs are in Zone 1 and one is in Zone 2. They're spread across three networks, and no two VMs share both a zone and a network. Task 4 shows why that matters.
Tip: Select the **nic0** link in the Internal IP column of any VM to see its subnet, IP range, the firewall rules and routes that apply to it, and other network analysis.

Task 4. Test connectivity between networks

Here you find out whether being in the same zone or being in the same network decides if VMs can reach each other.
Before you start, predict the results. Which VMs do you expect mynet-us-vm to reach by external IP, and which by internal IP?

Ping external IP addresses

1. On the **VM instances** page, note the external IP addresses of mynet-notus-vm, managementnet-us-vm and privatenet-us-vm.
2. In Cloud Shell, connect to mynet-us-vm again:

```
gcloud compute ssh mynet-us-vm \
--zone=[ZONE_1] \
--tunnel-through-iap
```

3. Ping each external IP address in turn, replacing the placeholder each time:

```
ping -c 3 [EXTERNAL_IP]
```

Ping internal IP addresses

1. On the **VM instances** page, note the internal IP addresses of the same three VMs.

2. In the mynet-us-vm SSH session, ping each internal IP address in turn:

```
ping -c 3 [INTERNAL_IP]
```

Expected results

Target VM	Same network as mynet-us-vm?	Same zone?	External ping	Internal ping
mynet-notus-vm	Yes	No (different region)	Succeeds	Succeeds
managementnet-us-vm	No	Yes	Succeeds	Fails (100% packet loss)
privatenet-us-vm	No	Yes	Succeeds	Fails (100% packet loss)

What this shows: External pings work to every VM, because public access is governed only by the ICMP firewall rules you created. Internal pings work only within the same VPC network, even across continents, and fail between networks even when the VMs sit in the same zone.

Task 5. Review

- You deleted the default network and confirmed that a VM can't be created without a VPC network.
- You built mynetwork in auto mode with firewall rules and two VMs, tested connectivity, then converted it to custom mode as recommended for production.
- You built managementnet and privatenet in custom mode, using the console for one and gcloud for the other, and added firewall rules and VMs.
- You confirmed that VPC networks are isolated from each other: internal traffic between networks is blocked unless you set up something like VPC peering or a VPN.

Check your understanding

1. True or false: without a VPC network, you can't create VM instances, containers or App Engine applications.
Answer: True.
2. Which firewall rule lets mynet-us-vm ping mynet-notus-vm's external IP address?
Answer: mynetwork-allow-icmp.
3. From mynet-us-vm, which VMs can you ping using internal IP addresses?
Answer: Only mynet-notus-vm, because it's the only one in the same VPC network.
4. Why convert an auto mode network to custom mode?
Answer: So new subnets aren't created automatically as new regions open, which could overlap with your own subnets or static routes. Custom mode gives you full control and is recommended for production.